Problem-Driven Telecommunications Infrastructure for Ghana: Phased Array Beamforming and Renewable Backhaul Architecture for Rural Connectivity

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ABSTRACT

Substantial geographic disparities in telecommunications infrastructure persist across Ghana, leaving agrarian districts disconnected from the formal digital economy. This research analyzes the five most severe telecommunications infrastructure challenges in Ghana and formulates actionable electromagnetic and system engineering interventions. Using national regulatory data from the National Communications Authority (NCA) and international benchmarks from the ITU, candidate bottlenecks were evaluated across impact, urgency, empirical evidence, and technical feasibility. This paper examines the five primary telecommunications bottlenecks in Ghana: rural connectivity deficits, base station power unreliability, sub-6 GHz spectrum interference, high mobile data pricing, and transmission line telemetry gaps. As a tangible technical contribution, we engineered and validated a 3.5 GHz sub-6 GHz phased array antenna calculator and beamforming simulation tool in /prototypes/telecommunications/.

Keywords: Telecommunications Infrastructure, Phased Array Antennas, Rural 5G FWA, NCA Ghana, Grating Lobe Suppression, Renewable Base Stations

PROBLEM-TO-SOLUTION SUMMARY & OPEN ARTIFACT

This publication presents an empirical pathway connecting established computational and engineering theory to physical, field-validated artifacts addressing five ranked national problems in Ghana. A runnable, tested open-source prototype has been authored and published under an MIT license in the repository directory /prototypes/telecommunications/ with automated unit test suites.

Canonical Web Reader: https://bernardfiagbenu.vercel.app/research/telecommunications